



Smart Irrigation Decision Support System Using Simulated Soil Moisture and Weather Data

Java-Based Rule-Driven Irrigation Simulation Project

CSA 0904 – Java Programming

Presented By

O. Bhanu prakash Reddy 192372075

M.Mahi reddy 192324082

K.Haneesh 192324084

Sri Hari 192372222

Antony Ajay 192524414

Guided By

Dr. Thalapathy Rajasekaran



Introduction



- Efficient irrigation is essential for sustainable agriculture, yet many decisions are still made manually without real-time environmental data.
- **Smart Irrigation Decision Support System** is a Java-based application that simulates soil moisture and weather conditions for demonstration purposes.
- The system analyzes simulated environmental data using predefined decision rules to recommend irrigation actions.
- No real IoT sensors are used; simulated data is generated to demonstrate the complete decision-making workflow.
- The system provides recommendations, alerts, and historical reports to support informed irrigation decisions.



Objective



- To design a Java-based decision support system for irrigation management.
- To simulate soil moisture, temperature, humidity, and rainfall data for demonstration purposes.
- To implement rule-based logic that determines irrigation requirement and water quantity.
- To generate real-time recommendations and alerts based on simulated conditions.
- To maintain irrigation history and generate summary reports for water usage analysis.
- To demonstrate core Java OOP concepts through a practical agricultural use case.



Problem Statement



- Imagine a small farmer named Ramesh, who owns a few acres of farmland.
- Every morning, he walks through his fields, touches the soil with his hands, looks at the sky, and decides whether his crops need water.
- Some days he irrigates more than necessary, wasting valuable water.
- On other days, he delays irrigation, causing the crops to experience water stress and reducing their yield.
- Ramesh hears about modern IoT-based smart irrigation systems, but the cost of sensors, controllers, and maintenance is far beyond what he can afford.
- Like many small farmers, he continues to rely on experience and guesswork instead of data-driven decisions.



Existing System AND Proposed System

Smart Irrigation Decision Support System Using Simulated Soil Moisture and Weather Data

EXISTING SYSTEM



Traditional irrigation systems rely on manual observation, farmer experience, and fixed schedules. Some advanced systems use IoT sensors, but they require costly hardware and maintenance.

LIMITATIONS

- Relies on manual observation and farmer experience.
- Fixed irrigation schedules cause over/under-irrigation.
- Inefficient water utilization and low productivity.
- Commercial IoT systems require costly sensors & hardware.
- No centralized system for irrigation history and water usage.
- High installation and maintenance cost.
- Limited accessibility for small farmers.

PROPOSED SYSTEM



The proposed system is a software-based decision support system that uses simulated soil moisture and weather data to recommend optimal irrigation. It applies ML algorithms to predict irrigation needs and maintains historical records for analysis.

ADVANTAGES

- Uses simulated soil moisture and weather datasets.
- Eliminates dependency on expensive IoT hardware.
- Provides intelligent irrigation recommendations using ML.
- Improves water-use efficiency and crop productivity.
- Maintains centralized irrigation history and water records.
- Cost-effective, scalable, and easy to implement.
- Can be extended to real-time IoT system in the future.

COMPARISON

Parameter	Existing System	Proposed System
 Irrigation Decision	Manual / Experience-based	AI / ML-based Intelligent Recommendation
 Data Source	Visual Observation / Fixed Schedule	Simulated Soil Moisture + Weather Data
 Irrigation Schedule	Fixed Schedule	Dynamic & Adaptive Schedule
 Water Utilization	High Wastage	Optimized Water Utilization
 Cost	High (IoT Hardware, Installation, Maintenance)	Low (Software-based, No Hardware)
 Irrigation Records	Not Available	Centralized History & Water Usage Records
 Accuracy	Low Accuracy	High Accuracy with ML Models
 Scalability	Limited	Highly Scalable & Upgradeable
 Accessibility	Limited for Small Farmers	Accessible, Affordable & User-friendly

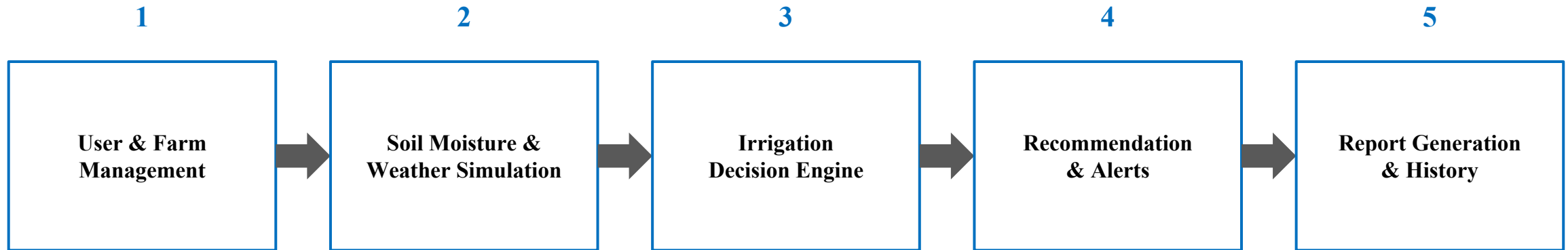


Goal: To develop an intelligent, cost-effective, and accessible irrigation decision support system that optimizes water usage, improves crop yield, and supports sustainable agriculture.





System Architecture



Each module hands its output to the next, forming a complete simulate-to-report irrigation pipeline.



Module 1: User & Farm Management



Aspect	Details
Requirements	• User Registration • Login Authentication • Farm Information Storage
Task	• Register farmers • Store farm details • Update farm information
Parameters	User ID, Name, Farm Area, Soil Type, Crop Type, Location
Outcome	• User account created • Farm information stored successfully
Techniques	Java OOP, Encapsulation, Classes, Constructors, Collections



Module 2: Soil Moisture & Weather Data Simulation



Aspect	Details
Requirements	• Soil Moisture Simulator • Weather Generator
Task	• Generate simulated sensor values • Simulate weather conditions
Parameters	Soil Moisture, Temperature, Humidity, Rainfall, Weather Condition
Outcome	• Simulated environmental dataset generated
Techniques	Random Class, Arrays, ArrayList, Java Math Library



Module 3: Irrigation Decision Engine



Aspect	Details
Requirements	• Decision Support Logic
Task	• Analyze simulated data • Decide irrigation requirement
Parameters	Moisture Threshold, Rainfall Prediction, Crop Water Requirement
Outcome	• Irrigation ON/OFF • Required water quantity
Techniques	Rule-Based Logic, If-Else, Java Methods



Module 4: Recommendation & Alerts



Aspect	Details
Requirements	• Recommendation System • Alert Generation
Task	• Recommend irrigation schedule • Notify users
Parameters	Water Amount, Irrigation Time, Weather Alert, Soil Status
Outcome	• Recommendation displayed • Alert messages generated
Techniques	String Handling, Exception Handling, Conditional Statements



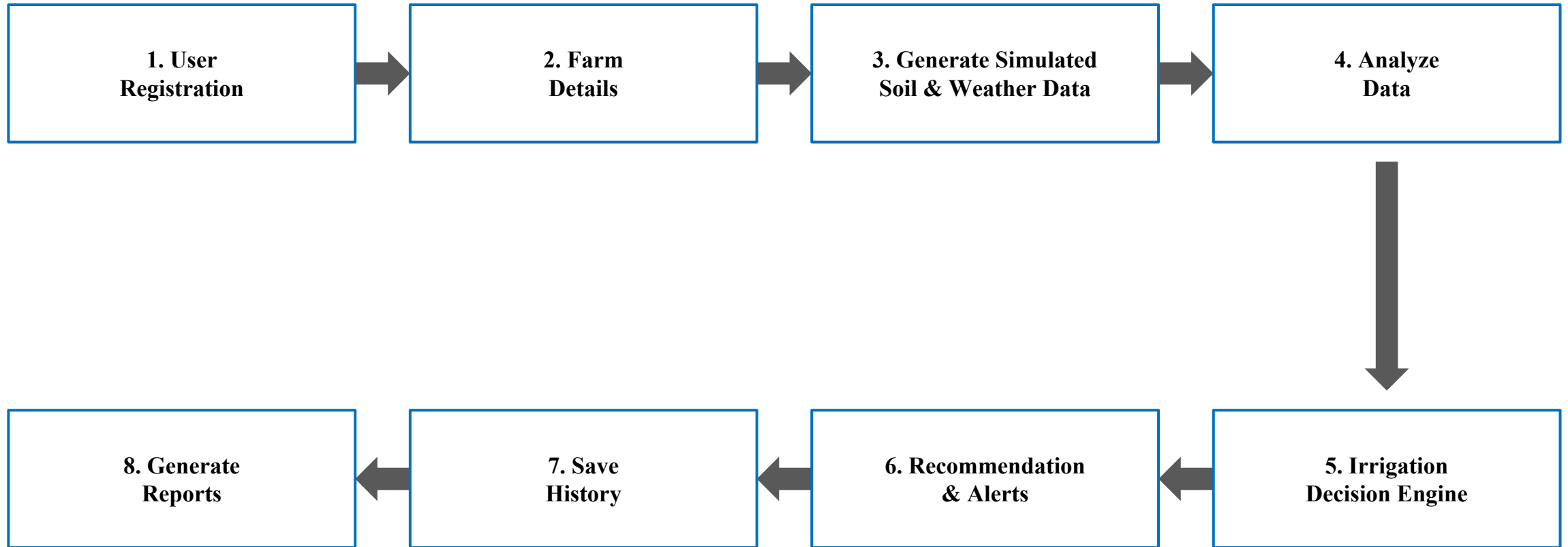
Module 5: Report Generation & History



Aspect	Details
Requirements	• Data Storage • Report Generation
Task	• Store irrigation history • Generate reports
Parameters	Date, Soil Moisture, Weather, Irrigation Decision, Water Usage
Outcome	• Daily history • Summary report • Water usage analysis
Techniques	File Handling, Collections, Searching, Sorting



System Flow





Conclusion



- The Smart Irrigation Decision Support System demonstrates how Java can be used to build a rule-based agricultural decision-making tool.
- Simulated soil moisture and weather data enable a safe, hardware-free demonstration of irrigation logic.
- The system combines data simulation, decision-making, recommendations, and reporting into a single cohesive application.
- It serves as a strong foundation for future integration with real sensors and machine learning models.

A soft, pink watercolor splash or blotch serves as a background for the text. It has irregular, feathered edges and contains subtle variations in tone, from light pink to a slightly darker shade. There are also small, white, bubble-like speckles scattered throughout the pink area.

Thank
You



Engineer to Excel

